PART D: DBR - Design Basis Report

Construction of 134 Residential Flats(2 Towers), DMD Bungalow, Office Building and Club house for SBI Enclave, Lingampally Post, Hyderabad in EPC Mode



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4 STRUCTURAL DESIGN BASIS REPORT

4.1 Introduction

The structure is designed for Residential Flats & Clubhouse, at Hyderabad. This report covers minimum design specifications, which will form the overall design philosophy to be adopted in the structural design of the project. The Structural Design Basis Report (DBR) is prepared in reference with the relevant IS Codes and Architectural drawings. 3D wire frame models of the building are generated and analyzed for various loads and load combinations using ETABS design software. The loads and load combinations were considered as per relevant IS codes. The RCC Building consists of

S.No	Block Name	Total Height (m)	Floor Height Clear (m)	No of Floors
1	Office	78.75	3.6	2B+ S + 20+(10 Future floors)
2	CGM & GM flats	74.1	3.2	B + S + 21
3	DGM flats	94.2	3.2	B + S + 27
4	Club house	18.75	3.6	B + G + 4

4.2 Site details

4.2.1 Project Location

The proposed project is located at Hyderabad.

4.2.2 Soil Data

Geo technical investigation is done and the report is enclosed as Annexure-1 (Geo-Technical Report)

The soil investigation report provided as part of the tender documents is for reference and bidding purposes only.

Any confirmatory soil testing or additional geotechnical investigations deemed necessary for design and execution shall be carried out by the EPC Contractor at their own cost, within the quoted contract price, and in full compliance with relevant IS codes and standards.

4.3 Structural system

4.3.1 Foundation

For this kind of proposed structure and sub soil condition raft foundation is the suitable foundation to adopt.
Depth of foundation should be, the embedded depth of the building shall be at least 1/15 of height of building for raft foundation and 1/20 of the height of building for pile and piled raft